

TANAGRA, Greece

WMO: 166990

Lat: 38.3353N

Lon: 23.5628E

Elev: 453

StdP: 14.46

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.8	32.1	19.8	15.1	37.1	24.4	18.8	37.8	22.9	52.7	20.1	52.4	4.5	300	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	97.0	68.6	94.6	68.3	91.6	67.9	72.4	89.5	70.8	87.6	69.7	86.1	9.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	102.1	78.1	65.9	97.0	76.4	64.3	91.8	75.4	36.2	89.0	34.9	87.6	33.9	86.1	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.4	14.2	DB	23.6	104.1	5.7	2.7	19.4	106.1	16.1	107.7	12.9	109.2	8.8	111.1	
			WB	22.9	74.7	5.2	2.7	19.2	76.7	16.2	78.3	13.2	79.8	9.5	81.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	45.7	47.4	51.6	57.8	66.1	74.6	79.4	79.2	72.0	63.6	55.4	48.1
	DBStd	13.11	6.52	6.32	5.95	4.91	5.17	5.11	3.84	3.61	4.96	5.14	5.97	6.29
	HDD50	458	165	110	52	4	0	0	0	0	0	0	17	110
	HDD65	2688	599	492	417	220	50	1	0	0	6	87	293	523
	CDD50	4772	31	38	101	238	499	738	911	904	660	420	180	52
	CDD65	1527	0	0	1	4	84	289	446	439	216	42	5	1
	CDH74	17527	0	0	14	110	968	3412	5567	5147	1979	312	18	0
	CDH80	8028	0	0	5	14	233	1523	2899	2568	740	45	1	0
(14) Wind	WSAvg	5.1	4.9	5.4	5.4	5.0	5.2	5.6	5.9	6.1	4.9	4.2	4.2	5.0
(15) Precipitation	PrecAvg	16.9	2.2	1.6	2.1	1.1	0.9	0.3	0.3	0.4	0.8	1.5	2.8	3.0
	PrecMax	26.3	5.2	4.1	7.0	2.4	2.1	0.7	1.3	1.1	3.9	5.6	5.9	6.3
	PrecMin	8.1	0.0	0.4	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.3
	PrecStd	4.5	1.6	0.9	1.7	0.6	0.7	0.2	0.4	0.4	1.1	1.5	1.6	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	68.4	74.7	82.2	89.5	100.5	103.9	100.0	95.4	84.5	76.7	69.6
		MCWB	54.8	55.4	56.0	60.1	65.9	67.5	69.1	69.2	66.4	64.4	61.2	59.2
	2%	DB	62.6	64.6	69.9	77.0	85.6	94.6	98.0	96.5	90.0	80.7	72.0	64.8
		MCWB	53.0	53.1	55.2	59.3	63.6	67.3	69.2	68.8	66.0	63.9	60.1	55.8
	5%	DB	60.5	61.4	66.6	73.4	82.4	91.2	94.7	93.4	87.4	77.3	69.5	62.6
		MCWB	52.3	52.1	53.7	57.8	62.1	67.1	68.5	68.4	65.5	62.9	59.8	54.1
	10%	DB	57.4	59.1	64.2	70.1	79.2	88.0	91.7	91.2	84.1	74.9	66.3	59.4
		MCWB	50.3	50.8	52.8	57.0	61.3	65.6	68.0	68.3	65.3	62.0	58.1	52.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.5	57.4	58.8	63.1	68.1	72.0	73.8	74.9	72.8	68.7	64.3	59.8
		MCDB	63.4	66.8	68.8	74.5	83.8	92.7	92.1	89.7	84.3	77.3	71.6	67.2
	2%	WB	54.5	54.7	56.7	60.9	65.8	69.8	72.1	72.7	70.0	66.6	62.0	57.2
		MCDB	61.0	61.9	66.6	72.1	80.6	87.6	90.2	89.6	81.1	75.0	69.2	63.7
	5%	WB	52.7	53.0	55.1	59.3	64.1	68.5	70.6	70.9	68.4	65.2	60.4	55.3
		MCDB	58.8	59.9	64.0	70.5	77.7	86.2	88.4	87.9	80.1	73.5	67.0	61.2
	10%	WB	50.8	51.5	53.5	57.9	62.7	67.2	69.5	69.6	67.1	63.6	58.9	53.1
		MCDB	56.1	57.8	62.5	68.9	75.8	84.3	87.2	86.2	79.1	71.9	65.0	58.9
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	16.9	19.0	21.6	22.5	23.1	22.8	21.9	20.5	18.0	16.3	14.9
		MCDBR	19.3	21.1	24.4	27.1	26.9	27.6	27.7	26.2	26.2	22.6	20.1	19.6
		MCWBR	12.2	12.6	13.0	13.0	10.7	9.7	9.6	9.1	9.9	10.5	10.8	12.4
	5% WB	MCDBR	17.8	19.4	21.9	24.2	24.1	24.8	23.6	23.2	21.3	19.4	18.0	16.7
		MCWBR	12.1	12.4	12.8	13.0	10.9	9.9	9.6	9.7	9.6	10.1	10.7	11.4
(40) Clear-Sky Solar Irradiance	taub		0.354	0.379	0.417	0.460	0.442	0.423	0.427	0.435	0.429	0.403	0.381	0.363
	taud		2.403	2.311	2.200	2.090	2.207	2.301	2.300	2.268	2.276	2.324	2.347	2.370
	Ebn at Noon		258	266	266	261	268	272	270	265	259	253	244	244
	Edh at Noon		28	34	42	50	45	41	41	42	39	34	29	27
(44) All-Sky Solar Radiation	RadAvg		592	836	1256	1683	2030	2328	2372	2108	1575	1057	705	514
	RadStd		78	108	110	161	108	128	61	54	96	99	78	76

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	-144	N/A	N/A

Nomenclature: See separate page